

Lecture Notes: Calculating Determinants

A practical mathematical introduction

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)}$$
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Row reduction $\implies$ triangular form
 $\implies$ product of diagonal entries

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1 What a determinant measures

A determinant is a number assigned to a square matrix. If

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

then its determinant is written as

$$\det(A) \quad \text{or} \quad \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.$$

Core interpretations

For a linear transformation represented by a matrix A , the determinant describes:

- the scaling factor of areas in $\mathbb{R}^2$ and volumes in $\mathbb{R}^3$;
- whether orientation is preserved or reversed;
- whether the matrix is invertible: A is invertible if and only if $\det(A) \neq 0$.

For this set of notes, the main goal is computational: how to calculate determinants efficiently and safely.

2 The determinant of a 2×2 matrix

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Example. A direct 2×2 calculation

Calculate

$$\begin{vmatrix} 3 & 5 \\ 2 & 7 \end{vmatrix}.$$

We use the rule $ad - bc$:

$$\begin{vmatrix} 3 & 5 \\ 2 & 7 \end{vmatrix} = 3 \cdot 7 - 5 \cdot 2 = 21 - 10 = 11.$$

Common mistake. Do not add the two products

The formula is $ad - bc$, not $ad + bc$. For example,

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2.$$

2.1 Geometric meaning in two dimensions

The absolute value $|\det(A)|$ is the area-scaling factor. For example, if

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix},$$

then A stretches the x -direction by 3 and the y -direction by 2, so it multiplies areas by 6. Indeed,

$$\det(A) = 3 \cdot 2 - 0 \cdot 0 = 6.$$

3 The determinant of a 3×3 matrix

There are several methods for a 3×3 determinant. The most common introductory method is expansion along a row or column. For

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

one compact formula is

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

The signs are important:

$$+ \quad - \quad +.$$

Example. A 3×3 determinant by the first row

Calculate

$$\begin{vmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{vmatrix}.$$

Expand along the first row:

$$\begin{aligned} \det(A) &= 2 \begin{vmatrix} -1 & 4 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 4 \\ 5 & 1 \end{vmatrix} + 3 \begin{vmatrix} 0 & -1 \\ 5 & 2 \end{vmatrix} \\ &= 2((-1) \cdot 1 - 4 \cdot 2) - 1(0 \cdot 1 - 4 \cdot 5) + 3(0 \cdot 2 - (-1) \cdot 5) \\ &= 2(-9) - 1(-20) + 3(5) \\ &= -18 + 20 + 15 = 17. \end{aligned}$$

3.1 Sarrus' rule

For 3×3 matrices only, one may also use Sarrus' rule:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

This rule is convenient for 3×3 matrices but does not generalize to larger matrices.

Common mistake. Sarrus' rule is not a general determinant algorithm

There is no analogous diagonal-copy rule for 4×4 determinants. For 4×4 and larger matrices, use row reduction or cofactor expansion strategically.

4 Minors, cofactors, and Laplace expansion

Let $A = (a_{ij})$ be an $n \times n$ matrix.

4.1 Minor

The minor M_{ij} is the determinant obtained by deleting row i and column j from A .

4.2 Cofactor

The cofactor C_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The sign pattern begins as

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix}.$$

4.3 Laplace expansion

Expansion along row i gives

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

Expansion along column j gives

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

Strategy for Laplace expansion

Choose a row or column with many zeros. Every zero eliminates one cofactor computation.

Example. A 4×4 determinant with many zeros

Calculate

$$A = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 3 & 4 & 0 & 0 \\ 0 & 5 & 6 & 0 \\ 0 & 0 & 7 & 8 \end{pmatrix}.$$

Expand along the first row:

$$\det(A) = 2 \begin{vmatrix} 4 & 0 & 0 \\ 5 & 6 & 0 \\ 0 & 7 & 8 \end{vmatrix} - 0(\cdots) + 0(\cdots) - 1 \begin{vmatrix} 3 & 4 & 0 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{vmatrix}.$$

Both remaining determinants are triangular:

$$\begin{vmatrix} 4 & 0 & 0 \\ 5 & 6 & 0 \\ 0 & 7 & 8 \end{vmatrix} = 4 \cdot 6 \cdot 8 = 192, \quad \begin{vmatrix} 3 & 4 & 0 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{vmatrix} = 3 \cdot 5 \cdot 7 = 105.$$

Hence

$$\det(A) = 2 \cdot 192 - 105 = 279.$$

5 Determinants and row operations

The most efficient method for larger determinants is usually row reduction. The key is to track how each row operation changes the determinant.

Row operation	Effect on the determinant
Swap two rows	Multiplies the determinant by -1
Multiply one row by a scalar k	Multiplies the determinant by k
Add a multiple of one row to another row	Does not change the determinant

Triangular matricesIf A is upper triangular or lower triangular, then

$$\det(A) = a_{11}a_{22} \cdots a_{nn}.$$

So row reduction aims to transform the matrix into triangular form.

5.1 Gaussian elimination method

To compute $\det(A)$:

1. Use row operations of type $R_i \leftarrow R_i + kR_j$ to create zeros below the diagonal.
2. If you swap rows, record a factor of -1 .

3. Avoid multiplying rows by constants unless you carefully record the factor.
4. Once the matrix is triangular, multiply the diagonal entries and adjust by the recorded factors.

Example. A determinant by elimination

Calculate

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 8 & 14 \end{pmatrix}.$$

Use determinant-preserving operations:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 3R_1.$$

Then

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 8 & 14 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 5 \end{pmatrix}.$$

Next use

$$R_3 \leftarrow R_3 - 2R_2,$$

so

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 2 & 5 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

No row swaps or row scalings were used, so the determinant is unchanged. The matrix is upper triangular, hence

$$\det(A) = 1 \cdot 1 \cdot 1 = 1.$$

Example. A determinant with a row swap

Calculate

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 3 & 4 \\ 2 & 1 & 5 \end{pmatrix}.$$

Swap R_1 and R_2 :

$$\begin{pmatrix} 0 & 2 & 1 \\ 1 & 3 & 4 \\ 2 & 1 & 5 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 3 & 4 \\ 0 & 2 & 1 \\ 2 & 1 & 5 \end{pmatrix}.$$

One row swap multiplies the determinant by -1 . Now eliminate:

$$R_3 \leftarrow R_3 - 2R_1,$$

so

$$\begin{pmatrix} 1 & 3 & 4 \\ 0 & 2 & 1 \\ 0 & -5 & -3 \end{pmatrix}.$$

Then

$$R_3 \leftarrow R_3 + \frac{5}{2}R_2,$$

giving

$$\begin{pmatrix} 1 & 3 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{1}{2} \end{pmatrix}.$$

The triangular determinant after the swap is

$$1 \cdot 2 \cdot \left(-\frac{1}{2}\right) = -1.$$

Since we performed one row swap at the beginning,

$$\det(A) = -(-1) = 1.$$

6 Important determinant properties

The following properties are both computational tools and conceptual facts.

6.1 Linearity in each row

If one row is multiplied by k , the determinant is multiplied by k . If a row is a sum of two row vectors, the determinant splits into a sum.

For example,

$$\det \begin{pmatrix} ka & kb \\ c & d \end{pmatrix} = k \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

6.2 Alternating property

If two rows are equal, then the determinant is zero. More generally, if one row is a scalar multiple of another row, then the determinant is zero.

Example. Detecting determinant zero without calculation

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 5 & 1 & 0 \end{pmatrix}.$$

The second row is twice the first row. Therefore the rows are linearly dependent and

$$\det(A) = 0.$$

6.3 Transpose

$$\det(A^T) = \det(A).$$

This means every row property also has a corresponding column property.

6.4 Product rule

If A and B are square matrices of the same size, then

$$\det(AB) = \det(A) \det(B).$$

6.5 Inverse matrix

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

7 Choosing the right method

Situation	Recommended method
2×2 matrix	Use $ad - bc$ directly.
3×3 matrix	Use cofactor expansion or Sarrus' rule.
Many zeros in one row/column	Use Laplace expansion along that row/column.
Matrix is triangular or nearly triangular	Multiply diagonal entries after simple cleanup.
Large matrix with no special pattern	Use Gaussian elimination and track row swaps/scalings.
Two proportional rows/columns	Determinant is immediately zero.
Block triangular matrix	Multiply determinants of diagonal blocks.

8 Special structures

8.1 Diagonal matrices

For

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix},$$

we have

$$\det(D) = d_1 d_2 \cdots d_n.$$

8.2 Block triangular matrices

If

$$A = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix},$$

where B and D are square blocks, then

$$\det(A) = \det(B) \det(D).$$

8.3 Vandermonde determinant

The Vandermonde matrix is

$$V = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Its determinant is

$$\det(V) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

This is useful in polynomial interpolation and numerical analysis.

9 Determinants and systems of linear equations

Consider the linear system

$$Ax = b.$$

If $\det(A) \neq 0$, then the system has exactly one solution. If $\det(A) = 0$, then the system may have no solutions or infinitely many solutions.

9.1 Cramer's rule

For a system

$$Ax = b,$$

where A is invertible, the i -th component of the solution is

$$x_i = \frac{\det(A_i)}{\det(A)},$$

where A_i is obtained from A by replacing the i -th column by b .

Example. Cramer's rule in a 2×2 system

Solve

$$\begin{cases} 2x + y = 5, \\ 3x - 2y = 4. \end{cases}$$

Here

$$A = \begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ 4 \end{pmatrix}.$$

First,

$$\det(A) = 2(-2) - 1 \cdot 3 = -7.$$

Then

$$A_x = \begin{pmatrix} 5 & 1 \\ 4 & -2 \end{pmatrix}, \quad \det(A_x) = 5(-2) - 1 \cdot 4 = -14,$$

so

$$x = \frac{-14}{-7} = 2.$$

Also

$$A_y = \begin{pmatrix} 2 & 5 \\ 3 & 4 \end{pmatrix}, \quad \det(A_y) = 2 \cdot 4 - 5 \cdot 3 = -7,$$

so

$$y = \frac{-7}{-7} = 1.$$

Therefore the solution is

$$(x, y) = (2, 1).$$

Practical note

Cramer's rule is elegant and theoretically important, but it is usually not efficient for large systems. Numerical computation normally uses Gaussian elimination, LU decomposition, QR decomposition, or iterative methods.

10 Worked examples**Example 1: Determinant from a triangular matrix**

$$A = \begin{pmatrix} 4 & -1 & 2 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix}.$$

Since A is upper triangular,

$$\det(A) = 4 \cdot 3 \cdot (-2) = -24.$$

Example 2: A determinant with proportional columns

$$A = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 6 & 7 \\ 4 & 8 & 9 \end{pmatrix}.$$

The second column is twice the first column. Hence

$$\det(A) = 0.$$

Example 3: Elimination in a 4×4 determinant

Calculate

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 2 & 3 & 1 & 5 \\ 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}.$$

Use determinant-preserving row additions:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_4 \leftarrow R_4 - R_1.$$

Then

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & -1 & 1 & -1 \end{pmatrix}.$$

Next:

$$R_3 \leftarrow R_3 - R_2, \quad R_4 \leftarrow R_4 + R_2.$$

This gives

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}.$$

Next:

$$R_4 \leftarrow R_4 - 2R_3,$$

so

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The final matrix has a zero row, so its determinant is zero. Since we used only determinant-preserving row additions,

$$\det(A) = 0.$$

Example 4: Parameter value from determinant zero

Find t such that

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 0 & t & 3 \\ 2 & 1 & 4 \end{pmatrix} = 0.$$

Expand along the second row or use the first row formula:

$$\begin{aligned} \det(A) &= 1 \begin{vmatrix} t & 3 \\ 1 & 4 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 \\ 2 & 4 \end{vmatrix} + 1 \begin{vmatrix} 0 & t \\ 2 & 1 \end{vmatrix} \\ &= (4t - 3) - 2(0 - 6) + (0 - 2t) \\ &= 4t - 3 + 12 - 2t \\ &= 2t + 9. \end{aligned}$$

We need

$$2t + 9 = 0,$$

so

$$t = -\frac{9}{2}.$$

11 A reliable computation checklist

Before finalizing a determinant calculation, check the following:

1. Is the matrix square? Determinants are defined only for square matrices.
2. Are there proportional or equal rows/columns? If yes, the determinant is zero.
3. Is the matrix triangular? If yes, multiply diagonal entries.
4. If using cofactor expansion, did you use the correct sign pattern?
5. If using row reduction, did you record row swaps and row scalings?
6. Does the answer make sense? For example, a row of zeros must give determinant zero.

12 Exercises

Basic exercises

1. Compute

$$\begin{vmatrix} 4 & 7 \\ 2 & 5 \end{vmatrix}.$$

2. Compute

$$\begin{vmatrix} -1 & 3 \\ 6 & 2 \end{vmatrix}.$$

3. Compute

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{vmatrix}.$$

4. Compute

$$\begin{vmatrix} 2 & 0 & 1 \\ 3 & 1 & 4 \\ 5 & 2 & 0 \end{vmatrix}.$$

5. Decide without full calculation whether

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{pmatrix}$$

is zero.

Intermediate exercises

6. Compute using cofactor expansion:

$$\begin{vmatrix} 1 & 0 & 2 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 5 & 6 & 1 \\ 0 & 0 & 7 & 8 \end{vmatrix}.$$

7. Compute using row reduction:

$$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 8 & 5 \end{vmatrix}.$$

8. Find all t such that

$$\det \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix} = 0.$$

9. Find a such that

$$\det \begin{pmatrix} 1 & a & 0 \\ 2 & 3 & 1 \\ 0 & 4 & 2 \end{pmatrix} = 0.$$

10. Let A be a 3×3 matrix with $\det(A) = 5$. What is $\det(2A)$?

Challenge exercises

11. Compute

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}.$$

12. Suppose A and B are 4×4 matrices, $\det(A) = -2$, and $\det(B) = 3$. Compute $\det(AB)$ and $\det(A^{-1})$.
13. Compute

$$\det \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

14. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Compute $\det(A)$ and explain the result using linear dependence of rows.

15. Compute the determinant of the Vandermonde matrix for $x_1 = 1$, $x_2 = 3$, $x_3 = 4$:

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{vmatrix}.$$

13 Answers to exercises

1. $4 \cdot 5 - 7 \cdot 2 = 20 - 14 = 6$.
2. $(-1) \cdot 2 - 3 \cdot 6 = -2 - 18 = -20$.
3. The matrix is upper triangular, so the determinant is $1 \cdot 4 \cdot 6 = 24$.
4. Expanding along the first row:

$$2 \begin{vmatrix} 1 & 4 \\ 2 & 0 \end{vmatrix} + 1 \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} = 2(1 \cdot 0 - 4 \cdot 2) + (3 \cdot 2 - 1 \cdot 5) = -16 + 1 = -15.$$

5. Yes. The third row is twice the first row, so the determinant is zero.
6. Expanding along the first row:

$$\det = 1 \begin{vmatrix} 4 & 0 & 0 \\ 5 & 6 & 1 \\ 0 & 7 & 8 \end{vmatrix} + 2 \begin{vmatrix} 3 & 4 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 8 \end{vmatrix}.$$

The first minor equals $4(6 \cdot 8 - 1 \cdot 7) = 4(41) = 164$. The second minor is triangular and equals $3 \cdot 5 \cdot 8 = 120$. Thus

$$\det = 164 + 2 \cdot 120 = 404.$$

7. Use row reduction:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 8 & 5 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence the determinant is 0.

8. $t^2 - 4 = 0$, so $t = \pm 2$.

- 9.

$$\det = 1 \begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} - a \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} = 1(6 - 4) - a(4) = 2 - 4a.$$

Therefore $a = \frac{1}{2}$.

10. For a 3×3 matrix, $\det(2A) = 2^3 \det(A) = 8 \cdot 5 = 40$.

11. This is a Vandermonde determinant:

$$(2 - 1)(3 - 1)(3 - 2) = 1 \cdot 2 \cdot 1 = 2.$$

Direct calculation also gives 2.

12. $\det(AB) = \det(A) \det(B) = -6$ and $\det(A^{-1}) = -\frac{1}{2}$.

13. Upper triangular, so $\det = 2 \cdot 3 \cdot 4 \cdot 5 = 120$.

14. $\det(A) = 0$. The rows are linearly dependent because

$$R_3 = 2R_2 - R_1.$$

15. Vandermonde formula gives

$$(3 - 1)(4 - 1)(4 - 3) = 2 \cdot 3 \cdot 1 = 6.$$

14 Summary

The determinant is a compact number that encodes invertibility, geometric scaling, and orientation. For computation:

- use $ad - bc$ for 2×2 matrices;
- use cofactor expansion or Sarrus' rule for simple 3×3 matrices;
- use row reduction for larger matrices;
- track row swaps and row scalings carefully;
- look for zeros, proportional rows or columns, and triangular structure before doing long calculations.